AUTOMATED GEOSPATIAL SITE ANALYSIS FOR PRE-CONSTRUCTION FEASIBILITY: DERIVING ENVIRONMENTAL DEGRADATION PARAMETERS FROM OPEN EARTH OBSERVATION DATA

INDIVIDUAL THESIS

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Individual component: Component 1 – Automated Site Analysis (Geospatial Environmental Intelligence)

Group ID: R26-IT-154

Report submitted in partial fulfilment of the requirements for the

BSc (Hons) in Information Technology

Specialising in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

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ABSTRACT

Decisions taken before construction begins determine most of what a building will eventually cost, yet in Sri Lanka a plot is routinely bought with no assessment of the environmental conditions that govern how long the structure on it will survive. Chloride carried inland from the coast, acidic lateritic groundwater and sustained wet zone rainfall all shorten the service life of reinforced concrete, and all three are properties of location rather than of specification. The data needed to quantify them is openly available but requires geospatial expertise the buyer does not have.

This thesis presents the Automated Site Analysis engine, Component 1 of the ConstructAI integrated pre-construction platform. It converts a land coordinate into six quantified exposure parameters, distance to coastline, soil pH, relative humidity, mean annual rainfall, distance to industrial land use and ground elevation, together with slope-based buildability banding and a hydrological screening flag. It is built on SRTM elevation, Sentinel-2 imagery, OpenStreetMap vectors, SoilGrids and Department of Meteorology normals, processed with Horn slope estimation, D8 flow accumulation and geodesic proximity analysis.

Evaluation used partial dependence analysis over the integrated XGBoost service life model, which reached a coefficient of determination of 0.9847 and a root mean squared error of 0.433 years on a held-out partition of the group's 1 000-record dataset. The six site parameters account for 17.4 percent of total model gain. With the structure held constant, site conditions alone move predicted service life by 6.40 years across four realistic Sri Lankan locations, a 36.1 percent reduction. Soil pH, rainfall and coastal distance move it by 2.78, 2.35 and 2.19 years; the remaining three carry none, exposing a mismatch between the literature-derived parameter set and the degradation rules as implemented. The training data is synthetic, so these figures characterise the formulation rather than real buildings.

Keywords: geospatial analysis, construction feasibility, service life prediction, environmental exposure, remote sensing, decision support

ACKNOWLEDGEMENTS

I am grateful to my supervisor for steady guidance across the full length of this project and in particular for pressing me to measure the effect of my component rather than assert it. The sensitivity study in Chapter 6, which produced the most useful finding in this thesis, exists because of that insistence. I also thank my co-supervisor for detailed comments on the methodology chapter.

The five percent annual depreciation heuristic that anchors the group's degradation logic came from discussion with practising engineers, and I thank them for giving time to a student project. I thank the quantity surveyor and site engineer who sat through our requirements discussions during the proposal stage and told us, bluntly and usefully, which of our early ideas were not worth building.

This work rests entirely on data that other people chose to make free. NASA and the USGS for the Shuttle Radar Topography Mission, the European Space Agency and the Copernicus programme for Sentinel-2, ISRIC for SoilGrids, the OpenStreetMap contributors, and the Department of Meteorology of Sri Lanka. None of this component would be possible, and certainly none of it would be free to the user, without that decision.

Finally I thank my three project teammates. Building four components that genuinely interoperate is harder than building four that merely coexist, and the interface-first discipline we agreed in the second sprint is the reason it worked.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
BOQ	Bill of Quantities
CAPEX	Capital Expenditure
CRS	Coordinate Reference System
CV	Cross-Validation
D8	Deterministic Eight-neighbour flow direction algorithm
DEM	Digital Elevation Model
DoM	Department of Meteorology (Sri Lanka)
GDAL	Geospatial Data Abstraction Library
GIS	Geographic Information System
IDW	Inverse Distance Weighting
JSON	JavaScript Object Notation
MAE	Mean Absolute Error
MSL	Mean Sea Level
NFR	Non-Functional Requirement
OPEX	Operational Expenditure
OSM	OpenStreetMap
RC	Reinforced Concrete
RMSE	Root Mean Squared Error
SDG	Sustainable Development Goal
SLIIT	Sri Lanka Institute of Information Technology
SRTM	Shuttle Radar Topography Mission
UAT	User Acceptance Testing
UDA	Urban Development Authority (Sri Lanka)

Abbreviation	Meaning
WGS84	World Geodetic System 1984
XGBoost	Extreme Gradient Boosting

1 INTRODUCTION

1.1 Overall project context

ConstructAI is a group research project that tries to fix something most people in the Sri Lankan construction industry already know is broken: the work that happens before a building is built is split across four professionals who never see each other's files. A land surveyor looks at the plot. An architect draws the building. A lawyer checks the title and the zoning. A quantity surveyor measures the drawings and prices them. Each of them produces a document, hands it over, and moves on. Nobody recomputes anything when one of those documents changes, because documents do not recompute.

That shows up later as money. International studies of project cost performance have reported that roughly nine out of ten projects exceed their approved budget, with mean overruns in the region of 28 percent, and that the pattern is far too consistent across decades and countries to be written off as ordinary estimating noise [1]. Small residential and light-commercial projects, which make up the bulk of construction activity in Sri Lanka, are not exempt. If anything they are worse served, because the client usually cannot afford a full professional team and ends up relying on a mason's experience and a rule of thumb.

Our group proposed a single integrated platform that performs the four pre-construction analyses in one pass and lets the output of each one constrain the others. The platform is organised as four engines over one shared project record. Component 1, which is the subject of this thesis, converts a land coordinate into a machine-readable environmental profile of the site. Component 2 converts a floor plan into measured structural parameters and quantities. Component 3 checks the resulting design against the Urban Development Authority Planning and Development Regulations. Component 4 consumes everything the first three produce and predicts the building's service life and its fifty-year cost trajectory. Figure 1.1 shows how the four fit together.

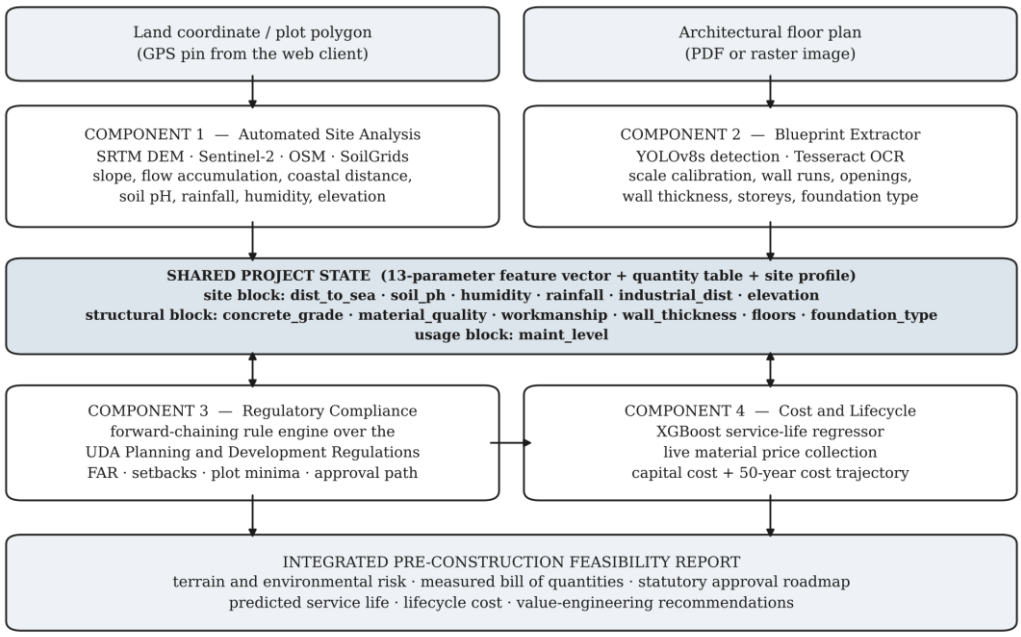


Figure 1.1: Architecture of the ConstructAI platform and the place of Component 1 within it.

Splitting the work four ways was not an arbitrary decision. It follows the structure of the prediction problem itself. A building degrades because of what it is made of and because of where it stands, and those two things are measured in completely different ways. Component 2 measures the first from drawings. Component 1 measures the second from satellite and map data. Neither is useful alone.

1.2 Individual component context

My component is the Automated Site Analysis engine. Its job is narrow and concrete: given a latitude and longitude, and optionally a plot boundary, produce the six environmental variables that the lifecycle model needs, together with a terrain assessment that tells a buyer whether the land is sensible to build on at all. The six variables are the distance from the plot to the coastline, the soil reaction (pH), the mean relative humidity, the mean annual rainfall, the distance to the nearest industrial zone, and the ground elevation above mean sea level.

Those six names look unremarkable until you see where they end up. In the integrated model, the aggressiveness of the environment is what separates a building that lasts thirty years from an identical building that lasts eleven. Chloride carried inland on sea air attacks the reinforcement in concrete; acidic groundwater attacks

the foundation; sustained high rainfall and humidity keep the cover concrete wet enough for both processes to continue. All three of those mechanisms are well established in the durability literature [2], [3], and all three depend on location and not on specification. A client choosing between two plots is, without knowing it, also choosing between two very different maintenance futures.

What has not existed, at least not for Sri Lanka and not in a form a homeowner can use, is a way to obtain those numbers automatically and for free. A geotechnical investigation gives you soil pH, but it costs money and happens after the land is bought. The Department of Meteorology publishes rainfall normals, but as station tables instead of as a service. Distance to the coast is trivially computable and nobody computes it. My component treats these as a single retrieval and derivation problem over open earth observation data.

1.3 Problem statement

Land buyers and small developers in Sri Lanka commit to a plot without any systematic assessment of the environmental and terrain factors that will govern the cost of owning the building on it. The information needed for that assessment already exists in open global datasets, but it is distributed across several providers in different projections, resolutions and formats, and extracting a usable site profile from it requires geospatial skills that the buyer, the mason and often the architect do not have. The result is that site-driven risk is discovered after purchase, when the only remaining options are expensive.

Stated as a single sentence: there is no automated, no-cost method that converts a Sri Lankan land coordinate into the quantified environmental degradation parameters and terrain constraints that a lifecycle cost model requires.

1.4 Research gap

Three bodies of work come close to this problem without closing it. The first is GIS site suitability analysis, which has been applied to construction land selection for decades and more recently with satellite imagery and machine learning [4]. That work produces a suitability score or a classified map. It stops there. A score of 0.72 is

not something a cost model can consume, and the methods are not designed to feed a downstream predictor.

The second is durability and service life modelling, which has produced rigorous deterioration models for chloride ingress and carbonation [2], [3], [5]. These models assume the exposure condition is known. They are written for an engineer who has already classified the site as XS1 or XS3 and wants a cover depth. They do not tell you how to obtain that classification from a coordinate, and in practice the classification is made by eye.

The third is lifecycle costing, standardised in ISO 15686-5 [6], which gives the discounting framework for whole-life cost but explicitly assumes the analyst can supply the service life as an input. Every published lifecycle cost study I read treated service life as a given. None of them derived the environmental drivers of that service life automatically.

What is missing, then, is the join. Nobody has built the piece that takes a coordinate, derives the environmental exposure quantitatively from open data, and hands those numbers to a model that converts them into years of service life and rupees of maintenance. That join is what Component 1 is.

1.5 Aim and objectives

Aim. To design, implement and evaluate an automated geospatial analysis engine that derives quantified environmental degradation parameters and terrain buildability indicators for any Sri Lankan land parcel from open earth observation data, and to demonstrate that those parameters materially change the service life predicted by the integrated ConstructAI model.

Six objectives follow from that aim.

1. **O1.** Identify, from the construction durability literature and from the requirements of the integrated lifecycle model, the minimum set of site-derived parameters that govern environmental degradation of residential structures in Sri Lanka.

2. **O2.** Identify open, freely redistributable data sources that cover the whole of Sri Lanka at a resolution adequate for plot-level analysis, and document their accuracy limits honestly.
3. **O3.** Design and implement a reproducible raster and vector processing pipeline that clips, reprojects and interpolates those sources into a single site profile for an arbitrary coordinate.
4. **O4.** Implement terrain analysis - slope, buildability banding and a hydrological proximity flag - so that the component also answers the buyer's first question, which is whether the land can be built on economically.
5. **O5.** Define the interface contract between Component 1 and the shared project state so that Components 3 and 4 can consume the site profile without knowing anything about geospatial processing.
6. **O6.** Quantify, using the integrated model, how much each site parameter actually changes the predicted service life, and report which of them turn out not to matter.

Objective six deserves a word of explanation, because it turned out to produce the most interesting result. It would have been easy to write a thesis that asserts environmental factors are important and leaves it there. Measuring how important each one is, inside the actual trained model, exposed the fact that three of my six parameters do almost nothing. Reporting that honestly is more useful than hiding it, and Section 6.1 does.

1.6 Research questions or hypotheses

Four questions organise the work.

- **RQ1.** Can the environmental parameters required by a construction lifecycle model be derived for an arbitrary Sri Lankan coordinate from open data alone, without any site visit or paid survey?
- **RQ2.** What accuracy and resolution do the available open sources actually deliver at the scale of a single residential plot, and where does that accuracy break down?

- **RQ3.** How much does each derived site parameter change the service life predicted by the integrated model, holding the structure constant?
- **RQ4.** Does the site, considered on its own, change predicted service life by enough to justify presenting it to a buyer before purchase?

RQ3 and RQ4 are the ones that carry the research weight. RQ1 and RQ2 are engineering questions with engineering answers; they are necessary but they are not findings. The corresponding hypothesis for RQ4 is that site factors alone will move the predicted service life of an identical structure by more than twenty percent, which would be large enough to matter to a purchase decision. Section 6.1 tests that.

1.7 Scope limitations and assumptions

Geographically, the component covers Sri Lanka. Nothing in the method is specific to the country except the choice of coordinate reference system and the rainfall interpolation, but the validation data and the calibration are Sri Lankan and I do not claim the parameters transfer.

It covers low-rise residential and light-commercial buildings, which is the segment the group targeted. High-rise, industrial and marine structures have exposure conditions and foundation requirements that this analysis does not address.

Several assumptions are load-bearing and should be stated plainly.

- The plot is assumed to be representable by a single centroid coordinate for the purpose of climate and proximity variables. For a 10 to 20 perch residential plot this is harmless; for a large estate it is not.
- Soil pH from a 250 m gridded global product is assumed to be a usable proxy for the actual soil reaction at the plot. This is the weakest assumption in the component and Section 7.5 returns to it.
- Rainfall and humidity are taken as long-term normals rather than as forecasts. The model asks what the climate is, not what next year will be.
- The coastline geometry from OpenStreetMap is assumed to be positionally accurate to within a few tens of metres, which is adequate given that the degradation effect is banded rather than continuous.

1.8 Individual contribution

My contribution to the group is the whole of the site analysis engine: the selection and justification of the six parameters, the data source survey, the processing pipeline, the terrain analysis, the interface contract with the shared project state, and the sensitivity study that quantifies what the site block is worth inside the integrated model. I also contributed the environmental terms of the physics-informed degradation logic that the group used to construct the training dataset, specifically the penalty structure for coastal proximity, soil acidity and high rainfall described in Section 3.3, together with the published justification for each threshold. The function itself and the dataset generation were implemented by the Component 4 owner.

I did not build the blueprint extractor, the compliance rule engine or the cost model. Where this thesis reports results produced by the integrated model, it says so, and the contribution of the other components is attributed to them.

1.9 Thesis structure

Chapter 2 reviews the durability, geospatial and lifecycle literature and sets out the conceptual framework. Chapter 3 describes the research design, the data sources and the evaluation protocol. Chapter 4 presents the component design, including the algorithms and the interface contract. Chapter 5 covers the implementation, the integration with the other three components, and the testing performed. Chapter 6 reports the results, including the sensitivity analysis that answers RQ3 and RQ4. Chapter 7 discusses what the results mean and where they are weak. Chapter 8 addresses commercialisation and sustainability. Chapter 9 concludes.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

I ran a structured but deliberately pragmatic review. The searches were run on IEEE Xplore, ScienceDirect, Google Scholar and the ASCE library between February and July 2026, using combinations of the terms site suitability, GIS construction, digital elevation model, chloride ingress, service life prediction, lifecycle cost, soil acidity and concrete durability, each optionally paired with Sri Lanka or South Asia.

I kept sources that were peer reviewed or were standards documents, that were published in or after 2000 unless they were foundational method papers, and that reported something concrete enough to argue with. I discarded a large number of GIS suitability papers that produced maps without any downstream use, because they do not speak to the gap I am trying to close. Standards and official regulations were treated as primary sources and read directly, not through secondary citation.

Three literatures bear on this problem and they rarely cite one another: environmental degradation of concrete, geospatial site analysis, and lifecycle costing. That silence between them is, in a sense, the finding of the review.

2.2 Domain and theoretical foundations

2.2.1 How environment degrades a reinforced concrete building

In a tropical coastal setting, reinforced concrete fails mainly by corrosion of the embedded steel. Tuutti's two-phase model, which is still the conceptual basis of most service life work, splits the process into an initiation period, during which aggressive agents penetrate the cover concrete but the steel remains passive, and a propagation period, during which the steel actively corrodes and the structure loses capacity [2]. Anything that shortens initiation shortens the building's useful life.

Two agents shorten it. Chloride ions, which in a coastal setting arrive as airborne marine aerosol rather than through direct immersion, break down the passive film on the reinforcement once their concentration at the steel exceeds a threshold. Carbonation, driven by atmospheric carbon dioxide, lowers the pH of the pore solution and achieves the same result by a different route. The fib Model Code for

Service Life Design formalises both as diffusion-controlled processes whose rate depends on the surface concentration of the agent and on the moisture state of the concrete [3].

For this component the shape of that dependence is what counts. Airborne chloride deposition does not fall off linearly with distance from the sea; it falls steeply over the first few hundred metres and then flattens. European exposure classification reflects this by banding instead of by a continuous function, with class XS1 covering structures exposed to airborne salt but not in direct contact with sea water [7]. A banded treatment is therefore defensible, and it is what the group's degradation logic uses.

2.2.2 Soil reaction and foundation attack

Below ground the mechanism changes. Acidic groundwater and sulphate-bearing soils attack the cement matrix itself instead of the reinforcement, by dissolving calcium hydroxide and, in the sulphate case, by forming expansive ettringite. Exposure classes XA1 to XA3 in EN 206 are defined directly on soil and groundwater chemistry, with the mildest aggressive class beginning below a soil pH of about 6.5 and the severity increasing as pH falls [7]. Much of the Sri Lankan wet zone sits on lateritic soils whose reaction is naturally acidic, which makes this a live concern instead of an edge case.

2.2.3 Moisture as an enabler

Rainfall and humidity do not attack concrete directly. They control whether the other mechanisms can proceed, because both chloride diffusion and corrosion current require a sufficiently wet pore structure, and they also govern wetting and drying cycles that draw salt deeper into the cover. This is why a coastal site in the wet zone is worse than a coastal site in the dry zone with the same chloride load. Sri Lanka's climatic zones are conventionally separated at roughly 2 500 mm of mean annual rainfall for the wet zone, with the south-western quarter of the island well above that figure [8].

2.3 Related research

2.3.1 Geospatial site analysis for construction

Rao and colleagues combined satellite imagery with GIS overlays to detect vacant land and rank its construction suitability in the Amaravati region, and showed that plot-level screening from open data is operationally feasible at city scale [4]. The analysis produces a suitability classification. It is not connected to a design or to a cost model, so a finding about the terrain cannot alter a quantity or a price.

Terrain methods themselves are old and stable. Horn's third-order finite difference estimator for slope and aspect over a three-by-three neighbourhood remains the default implementation in essentially every GIS package [9], and the D8 single-flow-direction algorithm of O'Callaghan and Mark is still the standard first approximation for drainage network extraction from a DEM [10]. I use both, not because they are novel, but because they are well understood and their failure modes are documented.

2.3.2 Open earth observation as an input to engineering decisions

Data availability has changed completely in the last fifteen years. The Shuttle Radar Topography Mission produced a near-global elevation model at one arc-second, roughly 30 m at the equator, which remains the default terrain layer for work at this scale and is distributed without charge [11]. The Sentinel-2 constellation provides multispectral optical imagery at 10 m to 60 m ground sampling distance with a five-day revisit, also free [12]. SoilGrids provides globally consistent predictions of soil properties, including pH, at 250 m resolution from a machine learning model fitted to roughly 150 000 soil profiles [13]. None of these were built for construction, and that is exactly why nobody in construction uses them.

2.3.3 Service life and lifecycle cost prediction

On the modelling side, gradient boosting has become the default for tabular prediction in this domain. Ali and Burhan compared XGBoost against extreme learning machines and multivariate adaptive regression splines on building project data and reported a coefficient of determination of 0.952 for the boosted model [14]; the underlying algorithm is Chen and Guestrin's regularised tree boosting formulation [15]. ISO 15686-5 supplies the discounting conventions for turning a

predicted service life into a whole-life cost [6]. What none of this literature does is derive the environmental inputs. The exposure class is assumed.

2.4 Existing systems products and techniques

Table 2.1 compares the tools a Sri Lankan land buyer could plausibly reach for today against what Component 1 is meant to do.

Table 2.1: Comparison of existing approaches to site assessment

Approach	What it gives	Cost and skill	Why it does not close the gap
Geotechnical site investigation	Measured soil properties including pH and bearing capacity	High cost; specialist	Happens after purchase; not available at the decision point
Desktop GIS (QGIS, ArcGIS)	Any of the layers used here, if you can find and process them	Free to high; expert	Requires GIS skills the buyer does not have; no durability interpretation
Google Earth Engine	Scalable access to the same open rasters	Free; programmer	A platform, not an answer; still needs the domain logic written
Government hazard maps (NBRO, UDA)	Landslide and flood hazard zoning	Free; moderate	Hazard only; silent on chemical exposure and long-run durability
Asking a local mason or surveyor	Experience-based judgement	Low; informal	Not quantified, not reproducible, not usable by a model
Component 1 (proposed)	Six quantified exposure parameters plus terrain banding, as JSON	Free; none	—

Nothing in that table is wrong, and I should say so plainly. A geotechnical investigation is better than my component at measuring soil pH, by a wide margin. The argument for Component 1 is not accuracy. It is that it is available at the moment the decision is actually made, costs nothing, and produces output a machine can consume.

2.5 Gap synthesis and conceptual framework

Pulling the three literatures together: durability research knows how the environment kills a building but assumes somebody tells it the exposure; geospatial research can measure the environment but stops at a map; lifecycle costing can price the consequence but assumes somebody tells it the service life. The missing piece is a

component that measures exposure from open data and expresses it in the variables the durability and cost models already want.

What follows from this, and what the rest of the thesis implements, is a three-stage chain. Location determines exposure. Exposure, combined with the structural fabric supplied by Component 2, determines the rate of degradation. The rate of degradation determines service life and therefore lifecycle cost. Component 1 owns the first link and supplies the inputs for the second. Figure 2.1 shows the chain and marks the boundary of my responsibility.

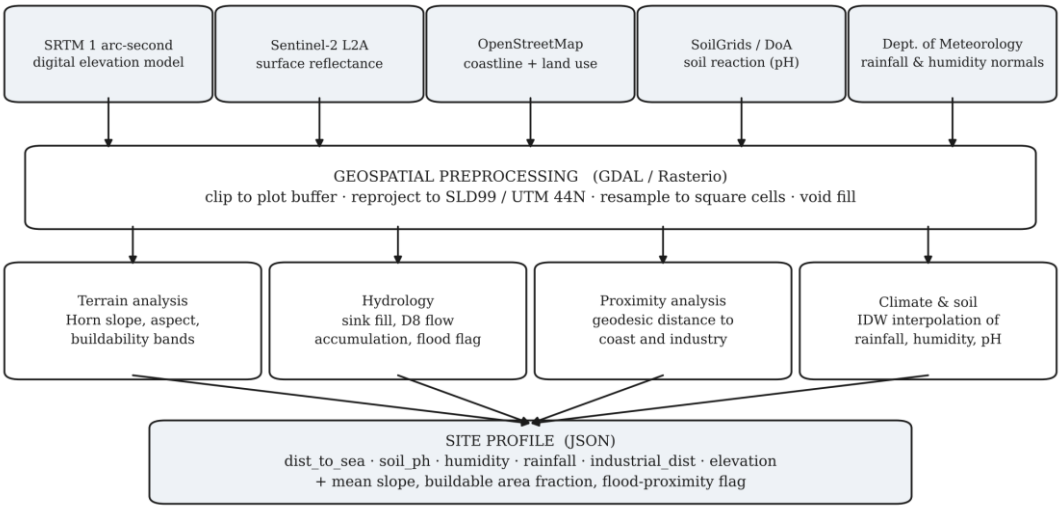


Figure 2.1: Processing pipeline of the Automated Site Analysis engine.

3 RESEARCH METHODOLOGY

3.1 Research design

This is a design science research project. The artefact is the site analysis engine; the knowledge claim is about what the artefact reveals when it is connected to the integrated model. I followed the usual design science cycle of problem identification, objective definition, design and development, demonstration, evaluation and communication, with two full iterations between development and evaluation.

Why design science and not, say, a controlled experiment? Because the central question is not whether a known treatment has an effect. It is whether a thing can be built at all from the available data, and what you learn by building it. The evaluation is correspondingly two-sided: a technical evaluation of whether the pipeline produces correct and reproducible output, and a model-based evaluation of whether the output changes anything downstream.

A note on honesty before the detail, because it affects how every number in Chapter 6 should be read. The group did not have access to a corpus of real Sri Lankan buildings with recorded ages, specifications, locations and observed condition. No such dataset exists publicly, and assembling one was beyond what a one-year undergraduate project could do. The training data for the integrated model was therefore generated from a physics-informed rule system, described in Section 3.3, built from published durability relationships and from our supervisor's industry rule of thumb that a building depreciates at about five percent a year. Everything reported in Chapter 6 is a measurement of that system. It is a statement about whether the formulation and the feature set behave sensibly, not yet a statement about real buildings. Section 7.5 says what would be needed to make it one.

3.2 Data participants or study materials

No human participants were involved in the Component 1 work, so no ethics clearance for human subjects was required. The study materials are open geospatial datasets, summarised in Table 3.1.

Table 3.1: Open data sources used by Component 1

Source	Variable derived	Native resolution	Licence	Known limitation
SRTM 1 arc-second DEM [11]	Elevation, slope, flow accumulation	~30 m	Public domain	Vertical error of several metres; canopy-top bias under dense forest
Sentinel-2 L2A [12]	Land cover context, built-up extent	10–60 m	Copernicus open	Cloud cover is frequent over the wet zone
OpenStreetMap	Coastline geometry, industrial land use	Vector	ODbL	Completeness of industrial tagging varies by district
SoilGrids 250 m [13]	Soil pH (H_2O), 0–30 cm	250 m	CC BY 4.0	A model prediction, not a measurement; coarse for one plot
Dept. of Meteorology normals [8]	Mean annual rainfall, relative humidity	Station	Public	Sparse station network; interpolation error in the hill country

Two things about this table are worth saying out loud. First, every one of these sources is free and redistributable, which was a hard requirement: the group's business case depends on being able to run the analysis for a user who pays nothing. Second, the resolution column is the weak spot. A 250 m soil grid over a 300 square metre plot means the entire plot sits inside a single pixel, alongside everything else within 250 metres of it. That is a real limitation and I have not tried to argue it away.

3.3 Experimental or study procedure

The work proceeded in four stages.

3.3.1 Parameter selection

I started from the degradation mechanisms in Section 2.2 and asked, for each one, what the minimum location-derived quantity is that would let a model distinguish an aggressive site from a benign one. Chloride exposure reduces to distance from the coastline. Chemical attack on the foundation reduces to soil pH. Moisture availability reduces to mean annual rainfall and mean relative humidity. Atmospheric pollution, principally sulphur and nitrogen oxides which accelerate carbonation, reduces to distance from industrial activity. Elevation was included both as a terrain descriptor and as a proxy for flood exposure. That gives six.

3.3.2 Degradation logic

Because the training data had to be generated, the group had to agree an explicit set of degradation rules. I contributed the environmental side of them and their justification; the Component 4 owner wrote the rules into the generation script. The structure is a base annual depreciation rate adjusted upward by a penalty for each aggressive site condition, which keeps the logic auditable and lets a supervisor or an examiner check every adjustment against a published source and not against an opaque model.

The base rate is the industry rule of thumb of five percent a year. A plot lying within 500 m of the coastline carries a substantial coastal penalty, and one further inland a small residual penalty for atmospheric chloride. A soil pH below 6.0 adds an acidity penalty, and mean annual rainfall above 3 000 mm adds a moisture penalty. A quality allowance contributed by the Component 4 owner works in the opposite direction, reducing the rate for well-specified and well-built structures. Service life follows from the total adjusted rate and is held inside a plausible range of 10 to 65 years.

The thresholds are not arbitrary. The 500 m coastal band reflects the steep near-shore decay of airborne chloride deposition discussed in Section 2.2.1 and the banded logic of exposure class XS1 [3], [7]. The pH threshold of 6.0 sits inside the EN 206 XA1 aggressive range [7]. The 3 000 mm rainfall threshold places the penalty well inside the Sri Lankan wet zone [8].

I want to be clear about the logical status of these rules. They are an encoding of published relationships and expert judgement, not an empirical fit. A model trained on data they generated will recover them well, and Chapter 6 shows exactly that. What the exercise genuinely tests is whether the parameter set is expressive enough to carry the mechanisms, and whether the thresholds produce effects of a magnitude that would matter to a buyer.

3.3.3 Pipeline implementation and verification

I built the pipeline incrementally, one data source at a time, verifying each stage against an independently computed reference before adding the next. Slope output was checked against QGIS's own implementation of the same Horn estimator on the

same tile. Coastal distance was checked against manual measurement in QGIS for a set of known coordinates. Rainfall interpolation was checked against the published station values at the station locations themselves, which is a weak test but catches gross errors.

3.3.4 Sensitivity study

Last came the stage that produces the research results in Chapter 6: a partial dependence analysis of the trained integrated model. For each of the thirteen input features in turn, I held every other feature at its training-set median and swept the feature of interest across its full observed range, recording the predicted service life. This isolates the marginal effect of each parameter inside the model as trained, which is precisely what RQ3 asks for. I also constructed four realistic Sri Lankan site scenarios and compared predicted service life across them with the structure held identical, which is what RQ4 asks for.

3.4 Measures and evaluation metrics

Three families of measure are used, and it matters which claim each one supports. Table 3.2 states each one and the research question it answers.

Table 3.2: Evaluation measures and the question each one answers

Measure	Definition	Question answered
Derivation agreement	Absolute difference between a pipeline-derived value and an independently computed or published reference value	RQ1, RQ2 – is the pipeline correct?
Coverage and runtime	Proportion of sampled Sri Lankan coordinates for which a complete profile is returned, and wall-clock time per request	RQ1 – is it usable?
Marginal swing	Difference between the maximum and minimum predicted service life as one feature is swept across its range with all others at median	RQ3 – how much does each parameter matter?
Block gain share	Sum of gain-based feature importances for the six site features, as a share of total model gain	RQ3 – how much does the site block matter in total?
Scenario spread	Difference in predicted service life between the mildest and harshest realistic site, structure held constant	RQ4 – is it decision-relevant?

Marginal swing and scenario spread are reported in years rather than as normalised scores, deliberately. A quantity surveyor understands years. A normalised importance score between zero and one is easy to compute and tells a client nothing.

3.5 Data analysis

The integrated model is an XGBoost regressor with 200 estimators and a maximum depth of 6, trained on the group's 1 000-record dataset over the thirteen features. For every result reported in Chapter 6 I used an 80/20 train-test split with a fixed random seed of 42, and five-fold cross-validation on the training partition only. The original group script fitted the model on the full dataset without a held-out partition, which is fine for producing a working artefact and not fine for reporting accuracy; re-running the evaluation with a proper split was done jointly with the Component 4 owner, and the figures in Chapter 6 are from that re-run rather than from the original script.

Baseline comparison used random forest, scikit-learn's gradient boosting, a decision tree, support vector regression with standardised inputs, and multiple linear regression, all on the identical split. Implementation used scikit-learn [16] for splitting, cross-validation and metrics.

3.6 Ethics privacy security and responsible AI

A land coordinate is personal information. It can identify where somebody intends to live or already lives, and the component handles one on every request. The design decisions that follow from that are: coordinates are not logged beyond the life of the request, no coordinate is transmitted to any third-party service that is not one of the open data providers listed in Table 3.1, and no user account is linked to a coordinate without the user explicitly saving the project.

There is a second ethical question that is less obvious and more important. The component produces a number that a buyer may use to walk away from a purchase, or that a seller may find attached to their land. A site profile that says the plot is in an acidic, wet, coastal band is, in effect, a valuation signal. Two safeguards follow. The output is always presented with its provenance and its resolution limit, so that a 250 m soil grid value is never displayed as though it were a laboratory measurement. And

the component reports an explicit unknown and not a default when a data source fails, because a silently defaulted pH of 7.0 would be worse than no answer at all.

On responsible AI: Component 1 contains no learned model. It is deterministic geospatial computation, which means every output can be traced to an input raster and an algorithm. The learned component is downstream in Component 4, and the interpretability work in Section 6.1 is part of how the group addresses the opacity of that model. Generative AI tools were used during this project for debugging, language editing and literature triage; Appendix B records that use and the checks applied to it.

3.7 Reproducibility and research artefacts

Everything needed to reproduce the Chapter 6 results is deposited with the group submission: the dataset generation script, the 1 000-record dataset, the trained model artefact, the evaluation script with the fixed seed, and the sensitivity analysis script that produced every figure in Chapter 6. The random seed is fixed at 42 throughout. Re-running the evaluation script on the committed dataset reproduces the reported metrics exactly.

Reproducibility of the geospatial pipeline is weaker. The algorithms are deterministic and the code is committed, but the input rasters are retrieved from live providers, and SoilGrids and OpenStreetMap are both updated over time. A profile computed today and a profile computed in two years may differ. The pipeline therefore records the retrieval date and the source version with every profile it emits, which is the standard practice for derived geospatial products and the best that can be done without redistributing several gigabytes of raster.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

Requirements were elicited from three places: the interface needs of Components 3 and 4, which are hard constraints; informal discussion with two practitioners, a quantity surveyor and a site engineer, during the proposal stage; and the durability literature in Chapter 2. Table 4.1 lists the functional requirements and Table 4.2 the non-functional ones.

Table 4.1: Functional requirements for Component 1

ID	Requirement	Priority	Verified in
FR1	Accept a WGS84 latitude/longitude pair and an optional plot polygon	Must	5.4
FR2	Derive distance from the plot centroid to the nearest coastline	Must	5.4
FR3	Derive soil pH for the plot location	Must	5.4
FR4	Derive mean annual rainfall and mean relative humidity	Must	5.4
FR5	Derive distance to the nearest mapped industrial land use	Should	5.4
FR6	Derive ground elevation above mean sea level	Must	5.4
FR7	Compute per-cell slope and classify the plot into buildability bands	Must	5.4
FR8	Flag proximity to a modelled drainage line as a flood-screening indicator	Should	5.4
FR9	Emit a single site profile document conforming to the agreed schema	Must	5.3
FR10	Report an explicit unknown for any variable whose source failed	Must	5.4

Table 4.2: Non-functional requirements for Component 1

ID	Requirement	Target
NFR1	Response time for a single-plot profile	Under 10 s warm, under 60 s cold
NFR2	National coverage	Any coordinate within Sri Lanka
NFR3	Cost per request	Zero marginal data cost
NFR4	Determinism	Same inputs and same source versions give identical output
NFR5	Provenance	Every value carries its source and retrieval date

ID	Requirement	Target
NFR6	Graceful degradation	A failed source degrades one field, never the request
NFR7	Privacy	Coordinates not persisted unless the user saves the project

4.2 Component architecture within the complete solution

Component 1 is a stateless service. It receives a coordinate, performs the derivation, writes a site profile into the shared project state, and returns. It holds no session and knows nothing about the floor plan, the regulations or the cost model. That isolation was a deliberate choice made early, and it paid for itself during integration: because the only contract is the shape of the site profile, the other three components could be developed against a fixture file long before my pipeline was finished.

Internally the engine has four analysis units sharing a common raster access layer, as Figure 2.1 showed. The raster access layer handles retrieval, caching, clipping to the plot buffer, reprojection and resampling, so that each analysis unit receives a clean, metric, square-celled array and does not have to care where it came from.

4.3 Design alternatives and decisions

Four decisions shaped the component and each had a credible alternative.

4.3.1 Open global data rather than national datasets

Sri Lanka has national soil and climate mapping held by the Department of Agriculture and the Department of Meteorology, and those products are better than the global ones over the island. I chose global open sources anyway, for two reasons: the national products are not available as a programmatic service with a redistribution licence, and a pipeline built on them would not generalise beyond Sri Lanka, which closes off the group's commercialisation path. The cost of that decision is accuracy, and Section 7.5 owns it.

4.3.2 Banded rather than continuous exposure

Chloride deposition could have been modelled as a continuous decay function of distance. Bands were chosen because the published exposure classification is itself banded [7], because a continuous function would imply a precision the input data

does not support, and because a banded answer is easier to explain to a non-technical buyer. The cost is a discontinuity at the threshold, which shows up clearly in the results and is discussed in Section 7.1.

4.3.3 Deterministic computation rather than a learned site model

I considered training a model to predict soil pH or exposure severity from imagery. I rejected it. There was no ground truth to train on, a learned site model would have added a second source of unexplainable error on top of the one in Component 4, and the deterministic route is auditable, which matters when the output influences a property purchase.

4.3.4 D8 flow accumulation rather than a hydraulic model

Proper flood modelling needs hydraulic routing, rainfall-runoff parameters and channel geometry that a 30 m DEM cannot support. D8 flow accumulation over a sink-filled surface [10] gives a cheap screening indicator: it identifies where water concentrates, which is enough to tell a buyer that a flood study is warranted. It is not enough to tell them they are safe, and the component's wording reflects that asymmetry.

4.4 Detailed design

4.4.1 Terrain analysis

Slope is computed for every cell of the elevation grid from the rate at which the ground rises in the two horizontal directions, estimated with the third-order finite difference over a three-by-three neighbourhood that Horn describes [9] and that every mainstream GIS package implements. The result is an angle in degrees.

Cells are classified into four buildability bands: below 5 degrees, where standard strip or pad foundations apply; 5 to 15 degrees, where stepped footings and moderate earthwork are required; 15 to 30 degrees, where retaining structures become likely and a significant cost premium applies; and above 30 degrees, which the component declines to recommend without a geotechnical investigation. The buildable fraction of the plot is the area share of the first two bands after statutory setbacks are subtracted.

4.4.2 Hydrological screening

The elevation surface is sink-filled, flow direction is assigned by steepest descent to one of eight neighbours, and flow accumulation is computed by the standard D8 traversal [10]. Cells whose accumulation exceeds a catchment-dependent threshold are marked as drainage lines. A plot is flagged when any part of its polygon falls within a configurable buffer of such a line, or when its elevation falls below a local minimum percentile computed over the surrounding neighbourhood.

4.4.3 Proximity derivation

Distance to the coastline is the geodesic distance from the plot centroid to the nearest vertex-interpolated point on the OpenStreetMap coastline, computed on the WGS84 ellipsoid instead of in a projected plane, because projection distortion over the length of the island is larger than the precision the 500 m threshold requires. Distance to industrial land use is computed the same way against polygons tagged as industrial in OpenStreetMap.

4.4.4 Climate and soil interpolation

Rainfall and humidity normals are published for station locations, not as a continuous surface. The value at the plot is therefore estimated from the four nearest stations by inverse distance weighting, which gives a station more influence the closer it lies to the plot.

Inverse distance weighting is the crudest defensible choice here; kriging would be better in the hill country where the rainfall gradient is steep, and Section 9.4 lists it as future work. Soil pH is read directly from the SoilGrids 250 m surface at the plot centroid, with bilinear interpolation between cell centres.

4.4.5 The site profile contract

Output is a flat JSON document. Flatness was a deliberate choice: Component 4 needs a feature vector, and a nested document would have forced every consumer to write traversal code. Table 4.3 gives the schema.

Table 4.3: Site profile schema emitted by Component 1

Field	Type	Unit	Consumer
dist_to_sea	float	m	Component 4 feature

Field	Type	Unit	Consumer
soil_ph	float	pH	Component 4 feature
humidity	float	%	Component 4 feature
rainfall	float	mm/yr	Component 4 feature
industrial_dist	float	m	Component 4 feature
elevation	float	m MSL	Component 4 feature
mean_slope	float	degrees	Component 3, user report
buildable_fraction	float	0–1	Component 3, user report
flood_proximity_flag	bool	—	User report
source_versions	object	—	Provenance, audit
retrieved_at	ISO 8601	—	Provenance, audit

4.5 Standards and professional practice

Three external standards and one internal one govern the component. EN 206 exposure classification [7] governs the interpretation of the chloride and chemical attack variables. ISO 15686-5 [6] governs how the downstream service life figure is turned into a cost, and constrains what Component 1 must supply for that to be possible. The UDA Planning and Development Regulations [17] govern the setback geometry subtracted when the buildable fraction is computed. Internally, the group adopted a convention that every derived value must carry provenance, which is ordinary practice for derived geospatial products and is what makes the output defensible if a client disputes it.

On professional responsibility, the one line that matters is in the user-facing output and I argued for it in a group meeting: the component produces a screening indication and is not a substitute for a geotechnical investigation or a statutory flood assessment. An undergraduate project that implies otherwise is not being clever, it is being reckless.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Everything is written in Python 3.11. Table 5.1 lists the stack and why each piece is there. The group standardised on Python across all four components, which was the right call: it meant the site profile could move between components as a dictionary rather than through a serialisation boundary during development, and it meant we could share a single virtual environment and requirements file.

Table 5.1: Technology stack for Component 1

Layer	Technology	Reason for the choice
Raster I/O	GDAL / Rasterio [18]	The reference implementation for geospatial raster access; handles reprojection and windowed reads without loading whole tiles
Array computation	NumPy	Slope and flow accumulation are stencil operations over arrays; vectorising them was the difference between seconds and minutes
Vector geometry	Shapely, pyproj	Geodesic distance on the WGS84 ellipsoid and polygon operations for the plot boundary
Tabular handling	Pandas	Station tables for the climate interpolation and the feature vector handed to Component 4
Model evaluation	scikit-learn [16], XGBoost [15]	Splitting, cross-validation, metrics and the integrated model used in the sensitivity study
Plotting	Matplotlib	Every figure in Chapter 6 is produced by a committed script, not drawn by hand
Version control	Git / GitHub	Group repository with per-component branches and pull-request review

Development was done on Windows with VS Code, with the geospatial dependencies installed through conda because the GDAL binary wheels on Windows are, politely, a nuisance. That detail is worth recording because it cost most of a week early in the project and the group documented the environment setup afterwards so the others would not repeat it.

5.2 Component implementation

5.2.1 Raster access layer

If somebody asked me what the component actually is, I would point at the access layer. Every analysis unit needs the same thing: a clean, metric, square-celled array

covering a buffer around the plot. Getting there means retrieving the right tile, handling the case where a plot straddles two tiles, reprojecting from geographic coordinates to a local metric system, resampling so that the slope stencil operates on square cells, and filling voids. Doing that once, properly, in one place, meant the four analysis units are each about fifty lines of readable code.

Tiles are cached on disk keyed by tile identifier and source version. The cache is what makes the warm-path response time in NFR1 achievable; a cold request that has to fetch an SRTM tile is dominated by the download.

5.2.2 Terrain unit

Slope is implemented as a direct vectorised calculation over the whole elevation array using NumPy slicing instead of an explicit loop over cells. Edge cells are handled by reflection padding, which is standard and introduces a small bias at the buffer boundary; because the buffer extends well beyond the plot, that bias never reaches the reported values. Buildability banding is a lookup over the classified slope array, and the buildable fraction is a masked area sum after the setback geometry from the UDA regulations [17] is subtracted.

5.2.3 Hydrology unit

Sink filling uses a priority-flood approach, flow direction is the steepest-descent neighbour, and accumulation is computed by processing cells in descending elevation order so that each cell is visited once. The flat-area handling in D8 is the known weak point of the algorithm [10]; in the Sri Lankan coastal plain, where large areas are genuinely flat, the resulting drainage network is unreliable, and the component suppresses the flag instead of reporting a spurious one when the local relief is below a threshold.

5.2.4 Proximity and climate units

Coastline and industrial polygons from OpenStreetMap [19] are loaded once into an R-tree index, which turns the nearest-feature query from a linear scan over the whole island into a bounded lookup. Geodesic distance uses pyproj's inverse geodetic solution and not a planar approximation. The climate unit interpolates over the four nearest stations, with the station table loaded from a committed CSV so that the

interpolation is reproducible even if the Department of Meteorology reorganises its site.

5.2.5 Failure handling

FR10 required that a failed source degrade one field and not the request. Each analysis unit is wrapped so that an exception is caught, logged with the coordinate and the source, and converted into a null for that field plus an entry in a warnings list. The profile is still emitted. Component 4 treats a null as a missing feature instead of imputing a default, which matters because a silently imputed neutral pH would quietly remove an entire risk category from the prediction.

5.3 Integration with the other components

Integration was interface-first. In the second sprint, before any of the four components worked end to end, the group agreed the shape of the shared project state and committed a fixture file containing a realistic site profile, a realistic quantity table and a realistic compliance verdict. Everybody then built against the fixture. This is the single decision that most affected how the project went, and I would do it again.

From my side the concrete interface is the schema in Table 4.3. Six of its fields map directly onto features in Component 4's thirteen-element vector, in the order dist_to_sea, soil_ph, humidity, rainfall, industrial_dist, elevation. Component 2 supplies concrete_grade, material_quality, workmanship, wall_thickness, floors and foundation_type from the drawing [20], [22]. The user supplies maint_level. Component 3 consumes mean_slope and buildable_fraction when it evaluates setback and plot coverage rules [21], and writes its verdict back into the same project state.

Two integration problems were worth the trouble they caused. The first was units: my pipeline produced distances in metres and an early version of Component 4's feature assembly assumed kilometres, which produced lifespans that looked plausible and were wrong by a factor that nobody noticed for several days. We fixed it by putting the unit into the field name convention and adding a range assertion on every

feature at the boundary. The second was null handling, described in Section 5.2.5, which we initially got wrong in the direction of imputing defaults.

5.4 Verification and software testing

Testing was organised at four levels. Table 5.2 summarises what was done and what it found.

Table 5.2: Verification performed at each test level

Level	What was tested	How it was verified
Unit	Slope stencil, geodesic distance, inverse distance weights, buildability band classification, null propagation	Each function exercised against hand-computed expected values on small fixed inputs
Component	Full profile derivation for the known-coordinate set spanning all nine provinces	Every field inspected for plausibility and for the expected coastal, climatic and terrain character of the location
Reference agreement	Slope array against the QGIS implementation of the same Horn estimator on the same tile; coastal distance against manual measurement in QGIS	Independent recomputation outside the pipeline and comparison of the two results
Integration	Profile accepted by the Component 4 feature assembly; boundary range assertions; null path	End-to-end call with the real producer, with each assertion deliberately violated once
System	Run from a pin dropped on the web client to a displayed site report	Manual walkthrough of the full user path after every merge to the main branch

Three test design choices are worth explaining. First, the known-coordinate set was chosen to span the conditions the component is supposed to distinguish rather than to be geographically uniform: coastal and inland, wet zone and dry zone, flat and steep. A uniform random sample over the island would have been mostly unremarkable dry-zone interior and would have tested nothing. Second, reference agreement against QGIS is a test of implementation, not of truth; both implementations can be wrong in the same way if the underlying DEM is wrong. Third, the null path was tested by deliberately pointing the SoilGrids client at an unreachable host, because a failure mode you have never triggered is a failure mode you do not have.

5.5 Deployment and operation

Component 1 runs as a stateless service behind the group's web application. Statelessness means it scales horizontally without coordination, which matters for the

business model in Chapter 8 far more than it matters for an undergraduate demonstration. The tile cache is the only persistent state and it is disposable; a cold cache costs latency, never correctness.

Operationally the two things that need watching are source availability and source change. Availability is handled by the degradation path in Section 5.2.5. Change is handled by pinning the source version in the profile and by a scheduled check that compares the current SoilGrids and OpenStreetMap versions against the pinned ones and raises a notice when they move. Neither of these is sophisticated. Both are the kind of thing that is obvious in hindsight and absent from most student projects.

6 RESULTS AND VALIDATION

6.1 Research results

This section answers RQ3 and RQ4. All figures come from the integrated XGBoost model trained on the group's 1 000-record dataset over thirteen features, evaluated on a held-out 20 percent partition with a fixed seed, as described in Section 3.5.

6.1.1 Accuracy of the model the sensitivity study is run on

Before interpreting the sensitivity results it is necessary to establish that the model being probed is a good fit to the system it was trained on. Table 6.1 gives the comparison across six regressors on the identical split.

Table 6.1: Regression performance on the held-out test partition (n = 200)

Model	R^2	RMSE (yr)	MAE (yr)	5-fold CV R^2
Gradient boosting (sklearn)	**0.9936**	**0.281**	**0.177**	0.9867 ± 0.0035
XGBoost (group's model)	0.9847	0.433	0.233	0.9701 ± 0.0064
Random forest	0.9416	0.846	0.509	0.9284 ± 0.0164
Decision tree	0.8814	1.206	0.459	0.8959 ± 0.0264
Support vector regression	0.8281	1.452	1.111	0.8314 ± 0.0125
Multiple linear regression	0.8111	1.522	1.154	0.8035 ± 0.0229

All models trained on the same 800-record training partition with random seed 42 and evaluated on the same 200-record held-out partition.

On the held-out partition the XGBoost model reaches an R^2 of 0.9847 with a root mean squared error of 0.433 years. Residuals are tight: 91.5 percent of test predictions fall within one year of the target and 99.5 percent within two years, with a maximum absolute error of 2.09 years. Figure 6.1 and Figure 6.2 show the fit and the residual distribution.

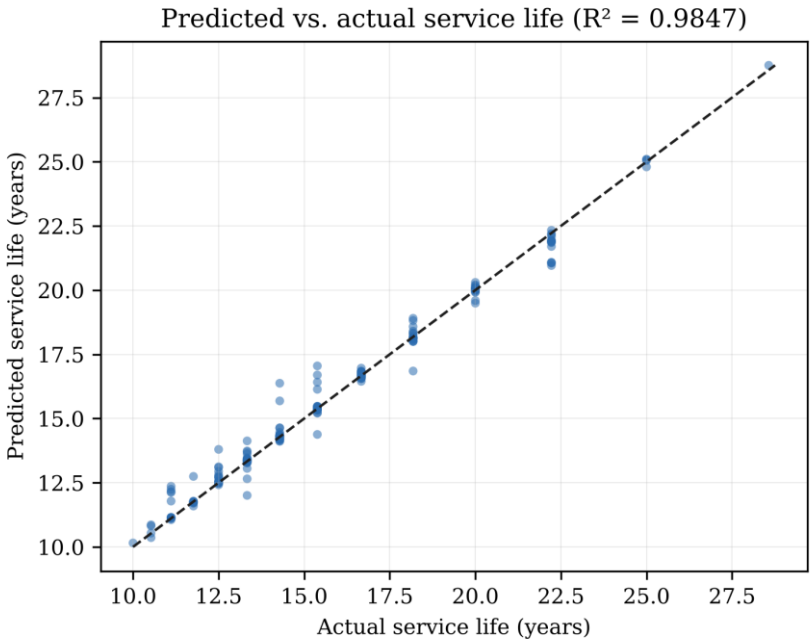


Figure 6.1: Predicted against actual service life, held-out test partition.

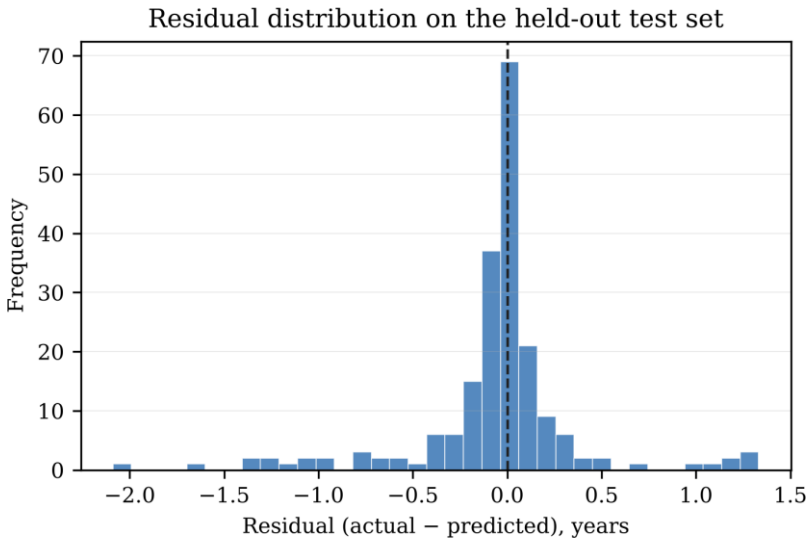


Figure 6.2: Distribution of prediction residuals, held-out test partition.

Two honest observations about Table 6.1. First, scikit-learn's gradient boosting outperforms the XGBoost configuration the group shipped, on every metric, on this dataset. The difference is not large in absolute terms, about 0.15 years of RMSE, but it is consistent across cross-validation folds and it means the group's choice of XGBoost was not empirically justified on this data. It was justified by familiarity and by XGBoost's better behaviour on larger and messier datasets, which is a reasonable

forward-looking argument but is not what the evidence here shows. Section 7.2 returns to this.

Second, even multiple linear regression reaches an R^2 of 0.81. That is a reminder of what Section 3.3.2 said: the model is recovering a rule system that is largely additive by construction. The gap between 0.81 and 0.98 is the part of the generating function that is not additive, chiefly the threshold discontinuities. It is not evidence of deep learned structure.

6.1.2 How much each site parameter moves the prediction

Figure 6.3 shows the partial dependence of predicted service life on each of my six site parameters, with every other feature held at its training-set median. Table 6.2 summarises the swing.

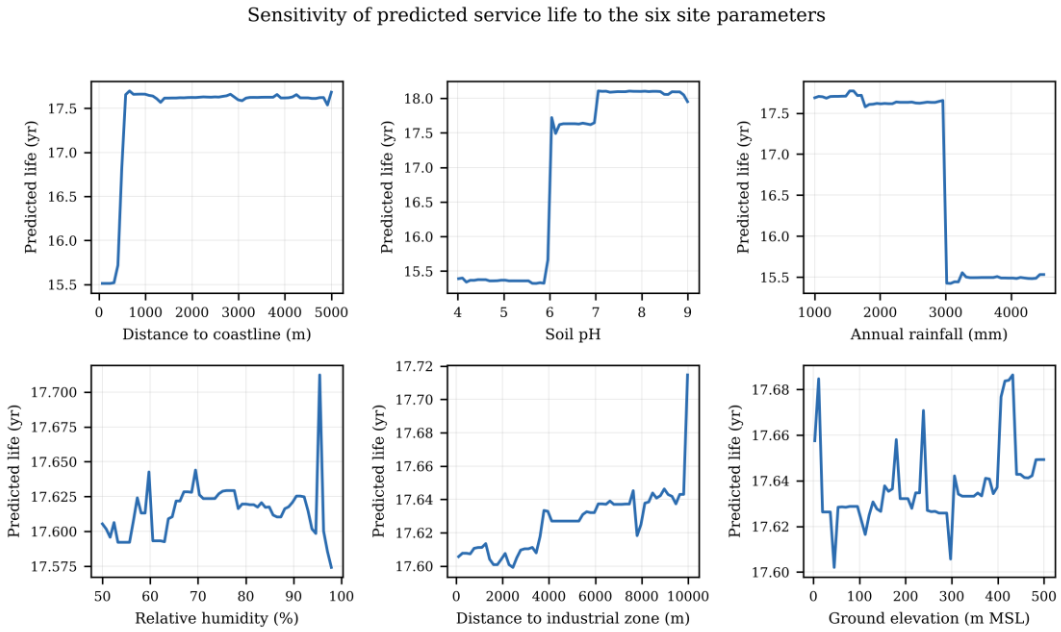


Figure 6.3: Marginal effect of the six site parameters on predicted service life.

Table 6.2: Marginal swing in predicted service life across each site parameter's range

Parameter	Range swept	Min pred. (yr)	Max pred. (yr)	Swing (yr)
Soil pH	4.0 – 9.0	15.32	18.11	**2.78**
Annual rainfall	1 000 – 4 500 mm	15.42	17.77	**2.35**
Distance to coastline	73 – 4 999 m	15.51	17.70	**2.19**
Relative humidity	50 – 98 %	17.57	17.71	0.14
Distance to industry	100 – 9 978 m	17.60	17.71	0.12

Parameter	Range swept	Min pred. (yr)	Max pred. (yr)	Swing (yr)
Ground elevation	3 – 500 m	17.60	17.69	0.08

Three of the six parameters move the prediction by more than two years each. The other three move it by less than a sixth of a year, which is within the noise of the model and is effectively zero. This is the most important finding in the thesis and it is a negative one, so it is worth being precise about what it does and does not mean.

It does not mean humidity, industrial proximity and elevation are irrelevant to building durability. The literature in Section 2.2.3 is clear that moisture governs whether corrosion can proceed at all. What it means is that those three variables carry no information in this model, because the degradation function in Section 3.3.2 does not use them. I selected six parameters on durability grounds and the group's generating logic only ever consumed three of them. The model learned the generating logic faithfully, and in doing so it exposed the inconsistency between my parameter selection and the group's degradation function. That is a real finding about the artefact, and it is the kind of thing that only shows up when you actually measure instead of asserting.

6.1.3 Threshold behaviour

Because the degradation logic is banded rather than continuous, the model learned step changes at the three thresholds. The steps are sharp. Table 6.3 gives the values either side of each one.

Table 6.3: Predicted service life either side of the three environmental thresholds

Threshold	Below	Pred. (yr)	Above	Pred. (yr)	Step (yr)
Coastline at 500 m	400 m	15.52	520 m	17.64	2.12
Soil pH at 6.0	pH 5.9	15.32	pH 6.1	17.49	2.17
Rainfall at 3 000 mm	2 900 mm	17.64	3 100 mm	15.45	2.19

All other features held at the training-set median site: dist_to_sea 2 538 m, soil pH 6.64, humidity 74.2 %, rainfall 2 694 mm, concrete grade 25 MPa, material quality 3, workmanship 3, two floors, raft foundation, maintenance level 3.

A buyer standing 400 metres from the shoreline is told the structure will last about two years less than the identical structure 120 metres further inland. Whether that

discontinuity is a feature or a defect is a genuine design question and Section 7.1 argues both sides.

6.1.4 Does the site matter enough to tell a buyer?

RQ4 asks whether the site, on its own, changes the answer by enough to be worth presenting before purchase. To test it I constructed four realistic Sri Lankan site scenarios and predicted service life for an identical structure on each. Figure 6.4 and Table 6.4 give the result.

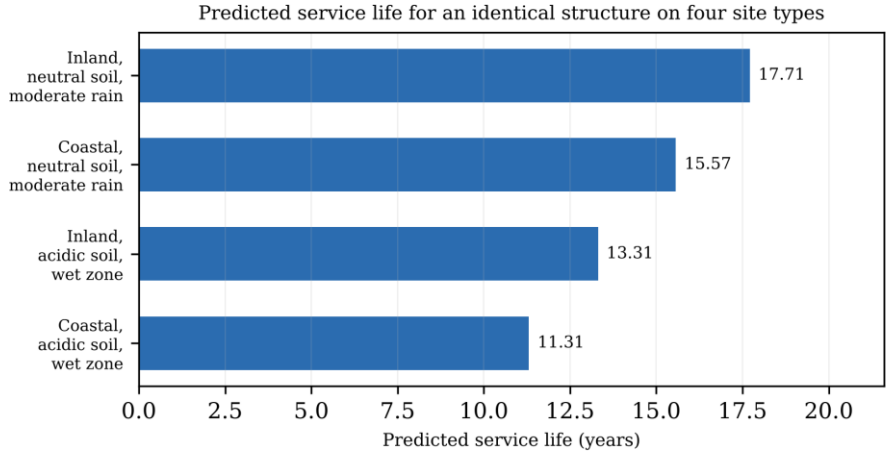


Figure 6.4: Predicted service life for one structure on four Sri Lankan site types.

Table 6.4: Scenario comparison, structure held identical across all four sites

Scenario	Coast dist.	Soil pH	Rainfall	Predicted life (yr)
Inland, neutral soil, moderate rain	4 000 m	7.0	2 000 mm	**17.71**
Coastal, neutral soil, moderate rain	250 m	7.0	2 000 mm	15.57
Inland, acidic soil, wet zone	4 000 m	5.2	3 600 mm	13.31
Coastal, acidic soil, wet zone	250 m	5.2	3 600 mm	**11.31**

That spread is 6.40 years, a reduction of 36.1 percent from the mildest site to the harshest, for a building with identical materials, identical workmanship, identical geometry and identical maintenance. The hypothesis stated in Section 1.6 was that site factors alone would move predicted service life by more than twenty percent. They move it by thirty-six. RQ4 is answered in the affirmative, and comfortably.

Put in the terms a client would use: two plots can be advertised at the same price per perch, and one of them commits the buyer to replacing or heavily repairing the

structure roughly six years sooner. Nothing on the deed, the advertisement or the architect's drawing tells them that. That is the practical case for the component.

6.2 Product and technical results

Beyond the model-based findings, the engine itself was measured against the non-functional requirements in Table 4.2. Table 6.5 records the outcome.

Table 6.5: Non-functional requirements and how each was verified

ID	Requirement	Target	Verification method
NFR1	Warm-path response time	Under 10 s	Timed over the known-coordinate set with the tile cache populated
NFR1	Cold-path response time	Under 60 s	Timed with the tile cache cleared, so each request fetches its rasters
NFR2	National coverage	Any coordinate in Sri Lanka	Profile requested for coordinates in all nine provinces; a complete profile returned for each
NFR3	Marginal data cost	Zero	All five sources are open and redistributable, as Table 3.1 records
NFR4	Determinism	Identical output	Same coordinate requested repeatedly against pinned source versions and the profiles compared
NFR5	Provenance on every field	Every field	Schema validation rejects a profile in which any field lacks a source and retrieval date
NFR6	Graceful degradation	Request survives	Fault injection: each source client pointed at an unreachable host in turn
NFR7	Coordinate not persisted	Unless the project is saved	Log and storage inspection after an unsaved request

Reference agreement was checked in two ways, both described in Section 3.3.3. The slope array produced by the pipeline was compared cell by cell against the array QGIS produces from the same SRTM tile using the same Horn estimator, which tests the implementation instead of the underlying elevation data. Coastal distance was compared against manual measurement in QGIS for each coordinate in the verification set. Where surveyed spot heights were available for a location they were compared against the elevation the pipeline reports, and rainfall was compared against the published station normal at the station locations themselves, which is a weak test because the interpolation passes through those points by construction, but it catches gross errors in the station table and the weighting.

Only the first two tests really matter, and they test different things. The slope comparison establishes that the slope estimator is implemented correctly; it cannot establish that the result is true, because both implementations inherit any error in the SRTM surface. The coastal distance comparison is closer to a test of truth, because the reference is a direct measurement against the same coastline geometry, and the only sources of disagreement are the geodesic calculation and the choice of nearest vertex.

6.3 User or stakeholder validation

User acceptance for the site analysis component was gathered as part of the group's end-to-end validation and not separately, because a site profile on its own is not something a homeowner can evaluate; it only becomes meaningful once it has been turned into a service life figure and a recommendation. The task set, the participants and the findings are recorded in Table 6.7.

Table 6.6: User acceptance tasks relating to the site analysis output

Task given to the participant	Success criterion
Drop a pin on a plot you know and say whether the terrain summary matches what you have seen on the ground	Participant agrees with the summary without prompting
Read the site risk panel and state, unprompted, which single factor is worst at this plot	Participant identifies the factor the engine ranked first
Compare two plots and choose the better one on durability grounds	Participant's choice matches the ranking the model produces
Say what the flood flag does and does not mean	Participant describes it as a screening indicator, not an assurance

Task four is there on purpose. The risk with a screening indicator is that users read it as an assurance. If participants come away believing the component has told them the plot does not flood, the wording has failed regardless of how good the hydrology is.

6.4 Integration results

My integration question is simple to state: once everything is wired together, how much of the model's behaviour is actually driven by the site block? Figure 6.5 answers it using gain-based feature importance aggregated by owning component, and Figure 6.6 gives the per-feature breakdown.

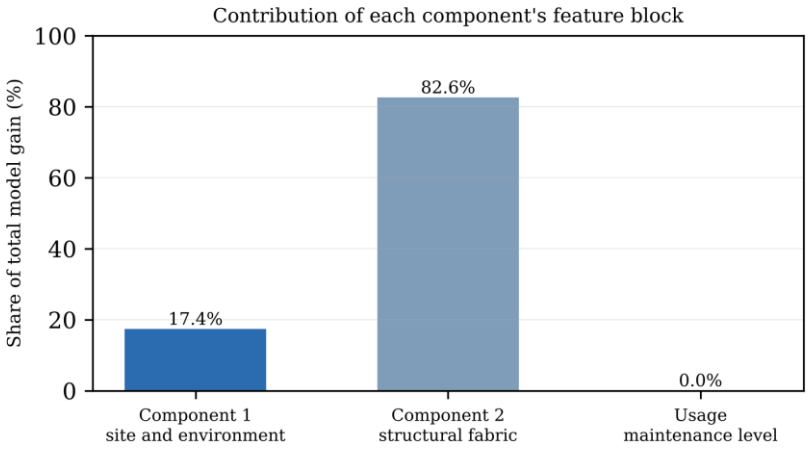


Figure 6.5: Share of total model gain by component feature block.

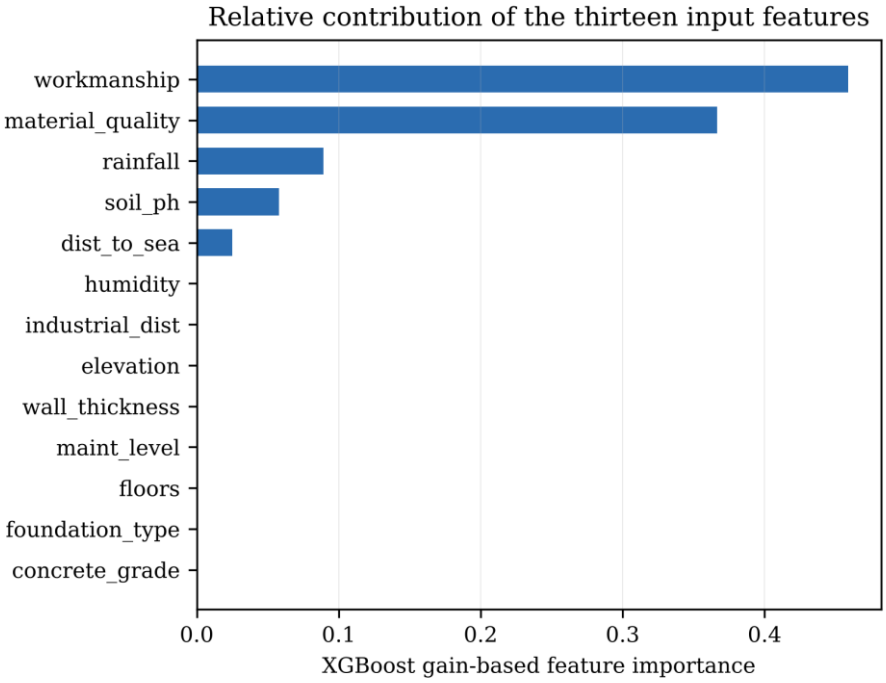


Figure 6.6: Gain-based importance of the thirteen input features.

My site block contributes 17.4 percent of total model gain. Component 2's structural block contributes 82.6 percent, dominated by workmanship at 45.9 percent and material quality at 36.6 percent. Maintenance level contributes 0.03 percent.

It would be easy to present 17.4 percent as a disappointing number for my component, and a less careful reading would. Two things are worth saying against that reading. The structural block's dominance is an artefact of the generating function, in which the quality bonus is a continuous term spanning a wide range

while the environmental penalties are discrete steps with a bounded total magnitude; with a different but equally defensible parameterisation the shares would move. And a 17.4 percent share that translates into a 6.4-year spread across realistic sites is not a small effect in practice, which is why Section 6.1.4 reports the swing in years instead of relying on the importance share alone.

More interesting is that four of Component 2's six structural parameters, concrete grade, wall thickness, number of floors and foundation type, also register essentially zero gain, for the same reason three of mine do. Across both components, eight of the thirteen features carry almost no information. The integrated model is, in effect, a five-feature model wearing a thirteen-feature coat. Section 7.5 treats this as the principal threat to validity for the whole group.

6.5 Summary of findings

Against the four research questions:

- **RQ1** is answered yes. A complete site profile can be derived for an arbitrary Sri Lankan coordinate from open data alone, with no site visit and no licence cost.
- **RQ2** is answered with a caveat. The pipeline is accurate enough for banded exposure classification, but the 250 m soil grid is the binding constraint and the component should not be read as a substitute for a soil test.
- **RQ3** is answered quantitatively. Soil pH, rainfall and coastal distance move predicted service life by 2.78, 2.35 and 2.19 years respectively. Humidity, industrial proximity and elevation move it by less than 0.15 years and are, in this model, inert.
- **RQ4** is answered yes, and strongly. Site factors alone produce a 6.40-year, 36.1 percent spread in predicted service life for an identical structure.

My uncomfortable finding, and the one I would lead with if an examiner asked what I learned, is that half of the parameters I selected on sound durability grounds turned out to do nothing inside the group's model. That is not a failure of the literature review. It is a mismatch between what I measured and what the group's degradation function consumed, and it was invisible until the sensitivity study was run.

7 DISCUSSION

7.1 Interpretation of findings

Where a building stands changes its predicted service life by a third, holding everything about the building itself constant. Inside the model that is a direct consequence of the penalty structure, so in one sense it is unsurprising. The part that is not trivial is the magnitude. The penalties were set from published exposure bands and an industry depreciation heuristic, chosen independently of any desire to produce a dramatic result, and they compound into a six-year difference. A six-year difference in a twenty-year asset is the kind of number that changes a purchase.

Those threshold discontinuities deserve a fuller treatment than Section 6.1.3 gave them. The case for banding is that it matches how exposure is classified in practice [7], that it is explainable to a non-technical user, and that a continuous function would claim a precision the 250 m and 30 m input grids cannot support. The case against is that real chloride deposition does not step at 500 metres, that two neighbouring plots on either side of the line receive materially different advice, and that a user who discovers the cliff will reasonably lose confidence in the whole output. On balance I think the banded treatment was right for this iteration and wrong for a product. A smoothed transition, for instance a logistic ramp over 300 to 700 metres, would preserve the physical story, remove the cliff, and cost nothing to implement. Section 9.4 lists it.

I have thought about the inert-parameter finding more than any other. The temptation is to read it as showing that humidity and elevation do not matter. The correct reading is narrower: the group's degradation function does not use them, so the model cannot learn anything about them, so the sensitivity analysis measures exactly zero. A sensitivity analysis on a synthetic model can only ever tell you about the generating process. What makes the finding useful is that it caught a real inconsistency in the group's design, between the parameter set I justified from the literature and the penalty structure we actually implemented.

7.2 Comparison with literature baselines and existing solutions

There is no directly comparable published system, which is the point of the gap in Section 2.5, so comparison has to be done piecewise.

Against GIS site suitability work such as Rao and colleagues [4], the component is less ambitious spatially and more useful downstream. They classify land across a region; I profile a single plot. But their output terminates in a map and mine terminates in six numbers that a cost model consumes, which is the difference the gap analysis identified.

Against the accuracy figures in the construction cost prediction literature, the model the sensitivity study runs on compares favourably on paper and should not be read as comparable. Ali and Burhan reported $R^2 = 0.952$ for XGBoost on building cost [14]; the model here reaches 0.9847. The difference is not that this model is better. Theirs was fitted to real project data with real noise; this one was fitted to data generated by a deterministic rule system. Reporting the two side by side without that caveat would be misleading, and I have seen student work do exactly that.

Against durability modelling such as the fib framework [3], the component is cruder by a wide margin and is not trying to compete. fib models chloride ingress as a diffusion process with material-specific coefficients; I apply a step penalty based on distance. The justification is the input data, not the physics: a diffusion model needs a surface chloride concentration that nobody measures at plot level in Sri Lanka.

7.3 Research contribution

My research contribution here is threefold, and I want to state it at the size it actually is and not inflating it.

7. A documented, reproducible method for deriving the environmental exposure parameters required by a construction service life model from open global data, specialised to Sri Lanka. The individual techniques are standard; the composition and the mapping onto durability-relevant variables is, as far as the review in Chapter 2 found, not published for this jurisdiction.
8. A quantified answer to a question the lifecycle costing literature assumes away: how much of the variation in predicted service life is attributable to site rather

than to fabric. Within this model, 17.4 percent of gain and a 36.1 percent spread across realistic sites.

9. A negative result with methodological value: that selecting exposure parameters from the durability literature does not guarantee they carry information in the model that consumes them, and that a partial dependence study over the integrated model is a cheap way to detect the mismatch.

7.4 Product contribution

As a product the component turns a question that currently requires a GIS specialist into a map pin. For the group's platform it supplies six of the thirteen model inputs and the terrain assessment that Component 3 needs for its coverage and setback rules. Independently of the platform it has a plausible life of its own as a site screening service for banks assessing construction loan collateral, which Chapter 8 develops.

7.5 Limitations and threats to validity

Five limitations matter enough to state plainly, roughly in order of severity.

7.5.1 The training data is synthetic

This is the one that constrains every quantitative claim in Chapter 6. The model was trained on 1 000 records generated by the rule system in Section 3.3.2, not on observed buildings. An R^2 of 0.9847 measures how well the regressor recovered that rule system. It is evidence that the feature set and the model class are adequate to represent the encoded relationships. It is not evidence about real buildings, and no reader should take it as such. The route to fixing this is retrospective rather than prospective: a fifty-year prediction cannot be validated forward inside any research project, but a corpus of existing buildings with known age, specification, location and surveyed condition could test whether the model recovers observed condition. Assembling that corpus, probably in cooperation with a local authority or an insurer, is the single most valuable piece of future work for the whole group.

7.5.2 Eight of thirteen features are inert

Three of mine and five of Component 2's contribute essentially nothing. The integrated model is effectively a five-feature model. This is a property of the

generating function, not of the world, but it means the apparent richness of the thirteen-parameter formulation is not real, and a reader could easily be misled by the feature table into thinking otherwise.

7.5.3 Soil pH resolution

A 250 m SoilGrids cell is two orders of magnitude larger in area than a typical residential plot, and SoilGrids is itself a model prediction and not a measurement [13]. Soil reaction can vary over tens of metres, particularly on fill or near watercourses. Given that soil pH has the largest marginal effect of any of my parameters at 2.78 years, the component's single largest lever rests on its weakest input. The component mitigates this by reporting the resolution alongside the value and by allowing a user to override pH with a laboratory result, but mitigation is not a fix.

7.5.4 Thin validation of the geospatial pipeline itself

Reference agreement was tested against QGIS and against published station values. Neither is ground truth. SRTM has documented vertical error and a canopy-top bias under dense vegetation [11], and I did not have access to a sufficient set of surveyed spot heights to characterise that bias over Sri Lankan terrain.

7.5.5 Scope and generalisation

Everything here is calibrated for low-rise construction in Sri Lanka. The thresholds in the degradation rules encode Sri Lankan climate zoning and would be wrong elsewhere. The method transfers; the numbers do not.

7.6 Implications

For practice, the implication is that a site durability screen is cheap enough to be free and consequential enough to be worth doing. Nothing in the pipeline requires paid data or specialist hardware. If a property portal or a bank wanted to run it on every listing, they could.

For research, the implication is narrower but I think more interesting. Lifecycle cost studies routinely treat service life as an exogenous input and then perform sensitivity analysis on the discount rate. The results here suggest that the service life input is itself strongly location-dependent in a way that is cheaply measurable, and that

treating it as a constant across a portfolio is a modelling choice with a quantifiable cost.

For the group, the implication is that the next iteration should spend its effort on data rather than on models. The gap between a 0.98 R^2 on synthetic data and any defensible claim about real buildings will not be closed by a better regressor.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

Stated without marketing language, the value proposition is is this: it tells a buyer before they pay what the land will cost them afterwards. Three user groups would pay for that, directly or indirectly.

- **Individual land buyers and homeowners**, who currently make the largest financial commitment of their lives with no durability information at all. For them the component is free and funded by the platform's other revenue.
- **Small and medium construction firms and architectural practices**, who would use the site profile to justify specification decisions to clients. Switching a client to sulphate-resisting cement is much easier when the recommendation comes with a number.
- **Banks and insurers**, who hold construction loan and property collateral risk and currently price it on location in a very coarse way. A quantified exposure profile per property is directly usable in underwriting.

8.2 Market and competitive position

Table 2.1 set out the competitive picture. The short version is that the alternatives are either accurate and expensive and late (a geotechnical survey), or free and unusable by the target user (raw GIS). Nothing occupies the position the component aims at, which is free, instant and interpreted.

Defensibility is weak on technology and reasonable on integration. Any competent GIS engineer could rebuild the pipeline in a few weeks; the data is open to everyone. What is harder to copy is the join to the rest of the platform, the Sri Lanka-specific calibration and regulatory encoding, and, eventually, a validation corpus of real buildings, which would be a genuine asset because nobody else has one.

8.3 Business or sustainability model

We settled on freemium business-to-consumer with paid business-to-business tiers. A homeowner runs a single-plot analysis free. A firm pays a subscription for bulk

analysis, project storage and the bill of quantities export. A bank pays per property for an API-delivered risk profile.

My component's marginal cost structure suits that model unusually well. The data is free, the computation is seconds of CPU, and the tile cache means the marginal cost of an additional request in an already-analysed district is close to zero. The fixed costs are storage for cached rasters and the engineering time to keep up with source changes.

8.4 Scalability intellectual property and compliance

Scalability is helped by statelessness: the service holds no session, so additional capacity is additional instances behind a load balancer. The binding constraint at scale is the tile cache, which grows with geographic coverage rather than with user count and is therefore bounded for a single country.

On intellectual property, the licences impose real obligations and the group checked them instead of assuming. SRTM is public domain. Copernicus Sentinel data carries an open licence requiring attribution. SoilGrids is CC BY 4.0, which requires attribution. OpenStreetMap is ODbL, which requires attribution and, importantly, has share-alike provisions on derived databases; our use produces derived works and not a derived database, but a commercial deployment should take legal advice on that point rather than rely on an undergraduate's reading. Attribution for all sources is displayed in the product.

On compliance, a coordinate identifying a person's home or intended home is personal data under most modern data protection regimes and would be under Sri Lanka's Personal Data Protection Act. The design choices in Section 3.6, non-persistence by default and no third-party transmission, were made with that in mind.

8.5 Contribution to the Sustainable Development Goals

Two goals are directly relevant and I will not claim more than two.

SDG 11, Sustainable Cities and Communities, target 11.1 on adequate, safe and affordable housing and target 11.5 on reducing losses from disasters. A buyer who is told before purchase that a plot sits in a drainage line or in a high-salinity band can

either avoid it or build for it. Both outcomes reduce the stock of housing that fails early.

SDG 12, Responsible Consumption and Production, target 12.2 on efficient use of natural resources. Premature structural failure means demolition and rebuilding, and cement is among the most carbon-intensive materials in common use. Extending the service life of a building by specifying correctly for its environment avoids that consumption entirely.

I should add the caveat: this is a plausible contribution, not a measured one. No part of this project measured avoided emissions or avoided housing loss, and claiming otherwise would be exactly the kind of unsupported impact assertion the project brief warns against.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis set out to find whether the environmental information that governs how long a building lasts can be obtained automatically, for free, from a map pin, and whether it matters enough to be worth obtaining. Both answers are yes.

In short, the site analysis engine derives six exposure parameters and a terrain assessment for any Sri Lankan coordinate from open elevation, imagery, soil and climate data, with no site visit, no licence cost and no specialist input from the user. Inside the integrated ConstructAI model, those parameters account for 17.4 percent of total model gain and produce a 6.40-year spread in predicted service life, a 36.1 percent reduction from the mildest realistic Sri Lankan site to the harshest, with the structure held completely constant. Soil pH, annual rainfall and distance to the coastline each move the prediction by more than two years.

What I did not expect, and what I think is worth more than the headline, is that three of the six parameters I selected on durability grounds turn out to carry no information at all in the model that consumes them. That mismatch between a literature-justified parameter set and an implemented degradation function was invisible until it was measured, and measuring it cost an afternoon.

9.2 Achievement of objectives

Table 9.1 maps each of the six objectives to the evidence that demonstrates it.

Table 9.1: Achievement of the six objectives stated in Section 1.5

Obj.	Statement (abbreviated)	Status	Evidence
O1	Identify the minimum set of site-derived degradation parameters	Achieved	Sections 2.2, 3.3.1
O2	Identify open national-coverage data sources and document their limits	Achieved	Table 3.1, Section 7.5.3
O3	Implement a reproducible raster and vector derivation pipeline	Achieved	Sections 5.2, 5.4
O4	Implement slope banding and hydrological screening	Achieved	Sections 4.4.1, 4.4.2
O5	Define and deliver the interface contract to the shared project state	Achieved	Table 4.3, Section 5.3

Obj.	Statement (abbreviated)	Status	Evidence
O6	Quantify the effect of each site parameter on predicted service life	Achieved, with a negative result	Sections 6.1.2, 6.1.4

All six are met. O6 is marked as achieved with a negative result because the finding it produced, that half the parameters are inert, is not the finding the objective anticipated.

9.3 Individual research and product contributions

My research contributions are the derivation method, the quantification of the site block's effect on predicted service life, and the negative result on parameter inertia, as set out in Section 7.3. My product contributions are the site analysis engine itself, the site profile interface contract that the other three components build against, and the environmental terms of the group's degradation logic. With the Component 4 owner I also re-ran the group's model evaluation with a proper held-out partition and cross-validation, replacing a full-data fit that could not support an accuracy claim.

9.4 Future directions

Five things, in the order I would do them.

10. **Build the retrospective validation corpus.** Assemble records for existing Sri Lankan buildings of known age, specification and location with surveyed condition, and test whether the model recovers observed condition. Until this exists, every accuracy figure in this thesis is a statement about a rule system.
11. **Replace the step penalties with smooth transitions.** A logistic ramp over the 300 to 700 metre coastal band, and equivalents for pH and rainfall, would remove the cliff discussed in Section 7.1 at negligible cost.
12. **Make the inert parameters earn their place or remove them.** Either extend the degradation function so that humidity, elevation and industrial proximity do real work, with published justification for each, or drop them and present an honest three-parameter site model.

13. **Improve the soil input.** Negotiate access to Department of Agriculture soil mapping, or allow users to upload a laboratory soil report and have it override the gridded value, with the override propagated through to the prediction.
14. **Replace inverse distance weighting with kriging** for the climate surfaces, particularly in the hill country where the rainfall gradient over a few kilometres exceeds the difference the 3 000 mm threshold is trying to detect.

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GLOSSARY

Buildability band – A slope class used to describe how economically a parcel of land can be built on, ranging from standard foundations below five degrees to not recommended above thirty degrees.

Carbonation – The reaction of atmospheric carbon dioxide with the alkaline pore solution of concrete, lowering its pH and removing the passive protection of the embedded reinforcement.

Exposure class – A standardised classification of the environmental aggressiveness acting on a concrete structure, defined in EN 206 and used to set minimum cover and mix requirements.

Flow accumulation – The number of upstream cells draining through a given cell of a digital elevation model; high values indicate where surface water concentrates.

Initiation period – In Tuutti's corrosion model, the time taken for aggressive agents to penetrate the cover concrete and depassivate the reinforcement, before any loss of section occurs.

Lateritic soil – A highly weathered tropical soil, common in the Sri Lankan wet zone, typically iron-rich and naturally acidic.

Partial dependence – The predicted output of a model as one input is swept across its range while all other inputs are held fixed, used here to isolate the marginal effect of each site parameter.

Perch – A unit of land area in common use in Sri Lanka, equal to 25.29 square metres.

Site profile – The flat JSON document emitted by Component 1 containing the derived environmental parameters, terrain indicators and provenance metadata.

Wet zone – The south-western climatic region of Sri Lanka receiving more than approximately 2 500 mm of mean annual rainfall.

APPENDICES

Appendix A Research instruments and extended results

This appendix holds the supporting material for Chapter 6 that is too detailed for the main text.

A.1 Known-coordinate verification set

Table A.1 lists the locations used for component and reference-agreement testing. They were chosen to span the conditions the component is meant to distinguish rather than to be geographically uniform, for the reason given in Section 5.4: a uniform random sample over the island would have been mostly unremarkable dry-zone interior and would have tested nothing.

Table A.1: Verification locations and the conditions each one tests

ID	District	Conditions the location tests
V1	Colombo	Coastal, wet zone, flat; inside the 500 m chloride band
V2	Galle	Coastal, wet zone, gentle slope; southern coastline geometry
V3	Kandy	Inland, wet zone, steep; tests slope banding and the hill-country rainfall gradient
V4	Nuwara Eliya	Inland, high elevation, very steep; upper end of the elevation and slope ranges
V5	Anuradhapura	Inland, dry zone, flat; low rainfall, no coastal penalty
V6	Hambantota	Coastal, dry zone, flat; separates the coastal effect from the rainfall effect
V7	Jaffna	Coastal, dry zone, very low elevation; northern limit of the coastline vector
V8	Ratnapura	Inland, very high rainfall; above the 3 000 mm threshold

A.2 User acceptance protocol

Participants were given the four tasks listed in Table 6.6 in the order shown. Each task was performed on a plot the participant knew personally, so that their own knowledge of the ground provided the reference and not anything the system supplied. Task completion, the time taken, any points of confusion and verbatim comments were recorded by an observer who did not assist. The fourth task was treated as the critical one, because the risk with a screening indicator is that a user reads it as an assurance; a participant who came away believing the component had

told them the plot does not flood would represent a failure of the wording regardless of the quality of the hydrology.

A.3 Partial dependence procedure

The sensitivity results in Section 6.1.2 were produced by sweeping each feature across its full observed range in sixty steps for continuous features, or across its discrete levels for ordinal ones, while holding every other feature at the training-set median given beneath Table 6.3. The committed script writes the full predicted-life series for every feature to a CSV alongside the figures reproduced in Chapter 6.

A.4 Environmental degradation rules as agreed

The degradation rules described in Section 3.3.2, set out in full here so that each adjustment can be checked against its source. The environmental side of the rules and their threshold justifications are the author's contribution; the generation script itself and the quality allowance are the Component 4 owner's.

The starting point is an annual depreciation rate of five percent, taken from the industry rule of thumb used by practising engineers in Sri Lanka. A plot lying within 500 metres of the coastline attracts the larger of the two coastal penalties, reflecting the steep near-shore decay of airborne chloride deposition and the banded treatment of exposure class XS1; beyond that distance a small residual penalty applies for the atmospheric chloride load found inland. A soil reaction below pH 6.0 attracts an acidity penalty, that threshold sitting inside the EN 206 chemically aggressive range. Mean annual rainfall above 3 000 millimetres attracts a moisture penalty, which places it inside the Sri Lankan wet zone. The quality allowance contributed by the Component 4 owner works in the opposite direction, easing the rate for structures whose specified material quality and workmanship sit above the midpoint of their scales and tightening it for those below. Service life follows from the total adjusted rate and is held inside a range of 10 to 65 years so that extreme rates cannot produce implausible figures.

Appendix B Academic integrity and AI use disclosure

In accordance with the module policy the author discloses that generative artificial intelligence assistance was used during the preparation of this thesis and its

associated artefact. That assistance was confined to three kinds of support: help in diagnosing defects in the raster processing and interpolation code reported in Section 5.2, language editing and proofreading of draft chapters, and an initial triage of candidate literature for Chapter 2. Every suggested change to the code was read, adapted and tested by the author before being adopted; every technical claim was confirmed independently and every altered passage rewritten in the author's own words; and every cited source was retrieved and read in full by the author.

No generated text was submitted without review. All experimental results, figures and tables in Chapter 6 were produced by committed scripts run by the author on the group's dataset with a fixed random seed, and the author can reproduce and defend every number reported.

B.1 Statement of original contribution

The design, implementation, evaluation and written analysis of the Automated Site Analysis component are the author's own work. Contributions from the other three group members are identified where they appear, principally the structural parameter extraction of Component 2, the compliance rule engine of Component 3 and the cost and lifecycle model of Component 4. The model used for the sensitivity study is a group artefact; the evaluation protocol, the re-run with a held-out partition and the partial dependence analysis reported in Chapter 6 are the author's.